NexusOS

Unified Enterprise Operating System

One Platform. Every Department. Connected Workflows.



United International University

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Al Amin Sawn
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Why NexusOS?

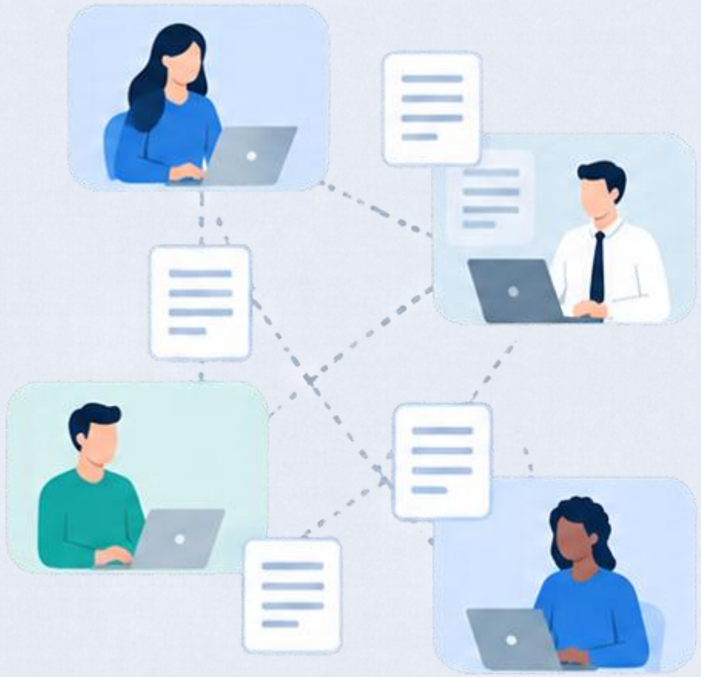
PROBLEM

Disconnected departmental systems.

Manual approval processes.

Scattered business information.

Difficult management and reporting.



SOLUTION

One centralized platform connects departments and automates business operations.



One Platform, Multiple Departments



HR & Employee Management



Finance & Accounting



Inventory Management



Procurement



Sales & CRM



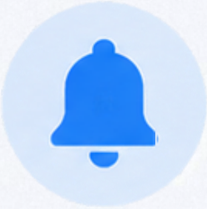
Project Management



Analytics & Reporting



Document Management



Notifications

One Request, Multiple Departments

1. IT Request — Purchase 5 Laptops
2. Manager Approval
3. Finance Approval
4. Procurement
5. Inventory Update



Workflow Automation Engine

1 TRIGGER
An event occurs.



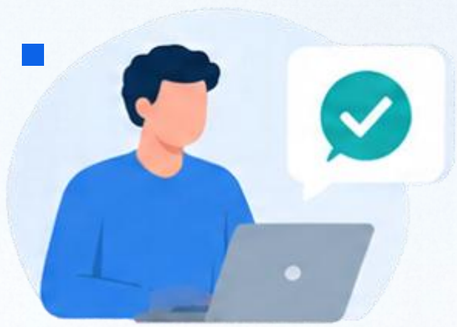
2 CONDITION
The system checks rules.

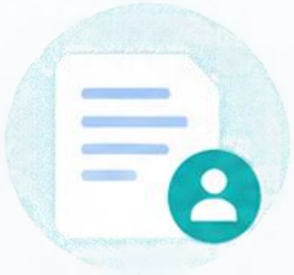


3 ACTION
The system performs a task.



4 APPROVAL
An authorized person approves.





Leave Request



Manager Approval

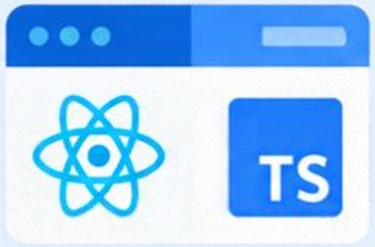


Attendance Update



Notification

How We Build NexusOS



Frontend

React +
TypeScript



Backend

Java + Spring Boot + REST
APIs



Database & Services

Supabase PostgreSQL + Supabase Storage
+ Firebase Authentication



Spring Security

for backend authorization



WebSocket

for real-time
communication

Built with Object-Oriented Programming

Encapsulation

Abstraction

Inheritance

Polymorphism

Design Patterns:

Strategy

Factory

Observer

State

Team Responsibilities



Member 1 — Frontend & UI



Member 2 — Backend & Workflow Engine



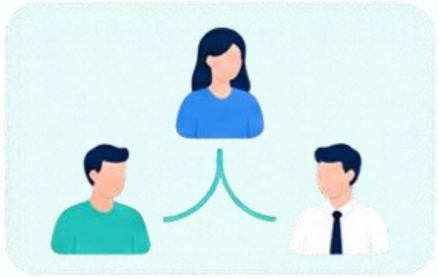
Member 3 — Database, Security & Testing

NexusOS

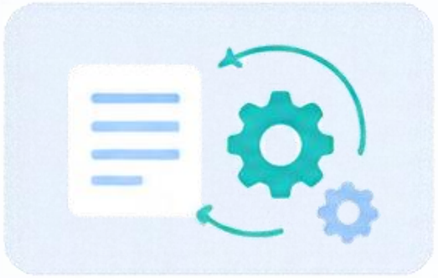
One Platform. Every Department. Connected Workflows.



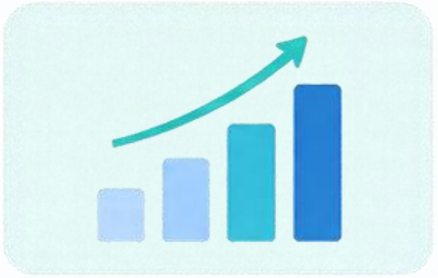
Centralized management.



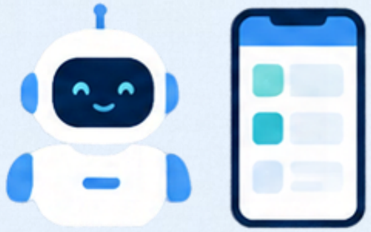
Connected departments.



Automated workflows.



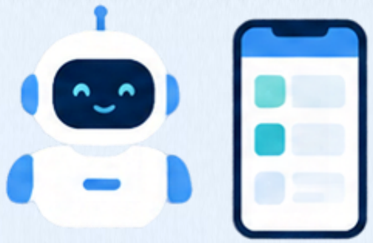
Better business visibility.



Future Scope:
AI Business Assistant and Mobile Application

THANK YOU

Questions & Answers



Future Scope:

AI Business Assistant and Mobile Application

THANK YOU

Any Questions??